

# SIE 433/533 Fundamentals of Data Science for Engineer

## Homework 2

**Due: February 20 (11:59 PM), 2025**

### Instruction

There are **3 (three) problems** in total. Each of them contains several sub-problems. You may use the knowledge learned from Python tutorial [Chapter 2](#), [Chapter 4](#), and [Chapter 5](#).

If you want to copy&paste your code and highlight the code, (1) go to <https://highlight.hohli.com/> (2) choose Python as the language (3) copy your code to the console, and (4) click *Highlight*.

### Submission Guidelines:

- Ensure to compile all your solutions into a **single PDF file** and submit it via D2L before the deadline.
- **Detailed intermediate steps are required for scoring.** Solutions that present only the final answer without supporting work will not be awarded any points.
- For problems solved using Python, **include the corresponding code within your PDF submission.** To aid in evaluating your work, please highlight the key segments of your code. This will streamline the review process, particularly in identifying any errors in your solution approach.

### Problem 1. (1.5 points)

A student defines three functions in Python, myfunc1, myfunc2, and myfunc3. Please answer the following questions.

```
1. def myfunc1(x, y):  
2.     z = x + y  
3.     print('%d' % z)  
4.  
5. result1 = myfunc1(1, 2)
```

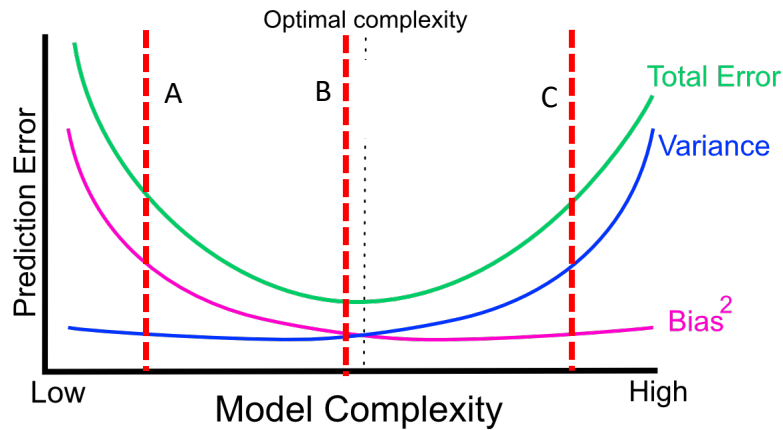
```
1. def myfunc2(x, y):  
2.     z = x + y  
3.     print('%d' % z)  
4.     return z  
5.  
6. result2 = myfunc2(1, 2)
```

```
1. def myfunc3(x, y=5):  
2.     z = x + y  
3.     print('%d' % z)  
4.     return z  
5.  
6. result3 = myfunc3(5, 3)
```

- 1) In myfunc1, what will be the result1? (0.5 points)
- 2) In myfunc2, what will be the result2? (0.5 points)
- 3) In myfunc3, what will be the result3? (0.5 points)

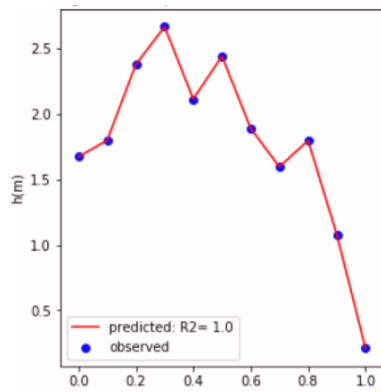
## Problem 2 (1.5 points)

In lecture 7, we have learned the bias-variance trade off as shown below,

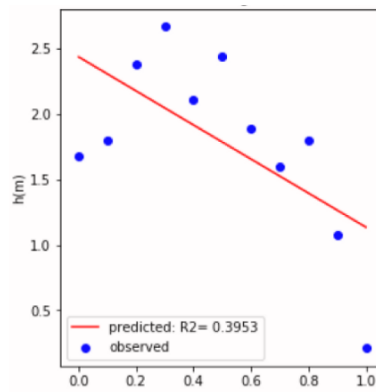


A student collected some data, and uses 3 different models to fit the data and draws the corresponding fitting plot (model prediction V.S true value).

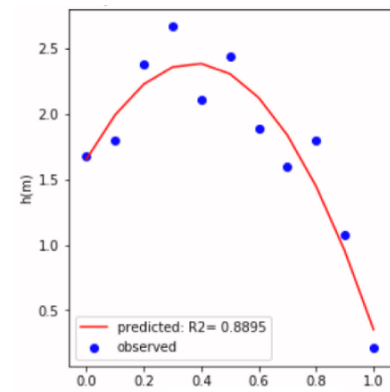
Please indicate which model complexity in the above figure (A, B, or C) are the following fitting models corresponding to. (simply write the answer, e.g., model X – A) (0.5 points for each matching pair)



Model 1



Model 2



Model 3

#### Problem 4 (7 points)

In this problem, you are asked to fit a linear regression model to the dataset 'Real\_estate\_valuation\_dataset.csv'. Set the y be the 'housePriceUnitArea' and the rest variables are the X. (Note: you do not need to use train\_test\_split(), please treat the whole dataset as the training dataset. Use default values for any unspecified parameters. You are allowed to use any functions)

- 1) Visualize the pairplot and correlation heatmap (1 point)
- 2) Fit a linear regression (with intercept) to the data and report the coefficients. (2 point)
- 3) Check the linear model assumptions by visualizing the plots. Does the model meet the assumptions? (2 points)
- 4) Use Python to calculate the  $R^2$  for models below and report the best model.  
Model-1: houseAge  
Model-2: houseAge, distanceToNearestMRTStation  
Model-3: houseAge, distanceToNearestMRTStation, numberOfConvenienceStores  
Model-4: houseAge, distanceToNearestMRTStation, numberOfConvenienceStores, latitude  
Model-5: houseAge, distanceToNearestMRTStation, numberOfConvenienceStores, latitude, longitude  
(1 point)
- 5) Use Python to calculate adjusted  $R^2$  for same models in subproblem-3, which more is the best model? (1 point)